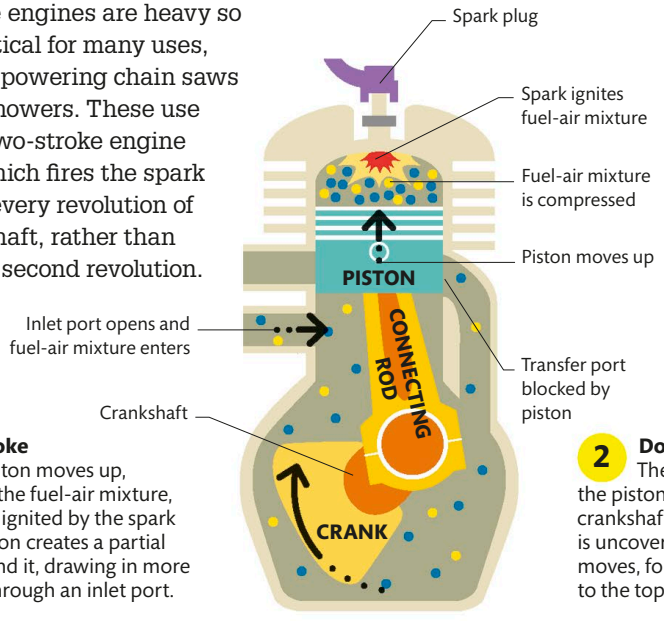
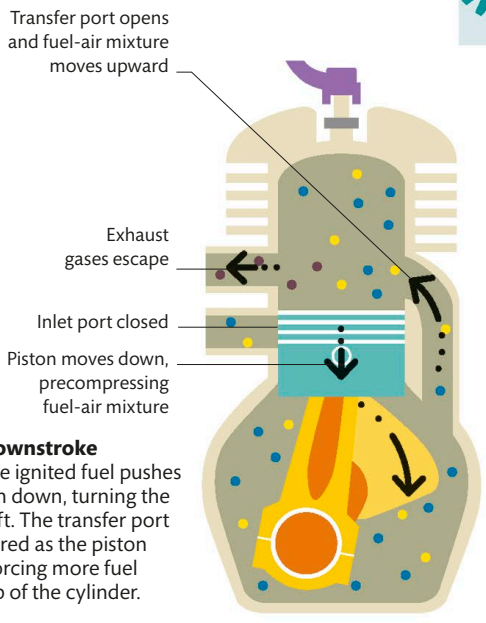
Two-stroke engines

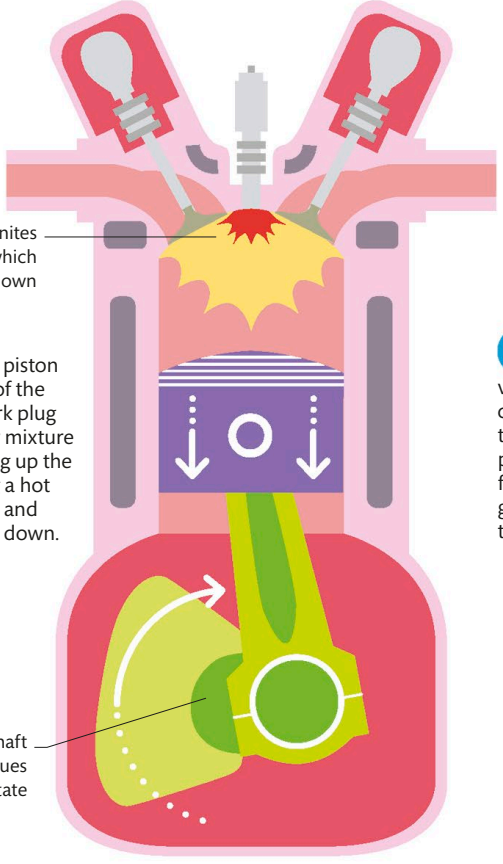
Four-stroke engines are heavy so are impractical for many uses, such as for powering chain saws and lawn mowers. These use a smaller two-stroke engine instead, which fires the spark plug once every revolution of the crankshaft, rather than once every second revolution.



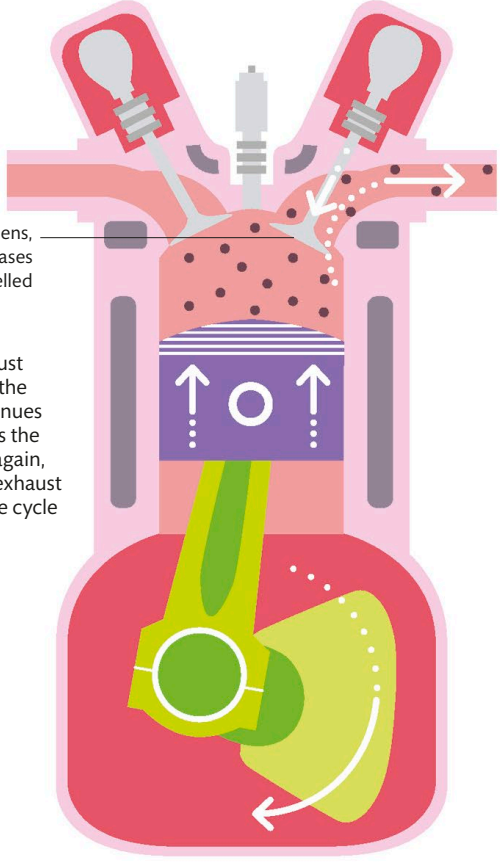
1 Upstroke
The piston moves up, compressing the fuel-air mixture, which is then ignited by the spark plug. The piston creates a partial vacuum behind it, drawing in more fuel and air through an inlet port.



2 Downstroke
The ignited fuel pushes the piston down, turning the crankshaft. The transfer port is uncovered as the piston moves, forcing more fuel to the top of the cylinder.



3 Power
When the piston reaches the top of the cylinder, the spark plug fires. The fuel-air mixture explodes, burning up the fuel and creating a hot gas that expands and forces the piston down.



4 Exhaust
The exhaust valve opens. As the crankshaft continues to turn, it pushes the piston back up again, forcing out the exhaust gases. The whole cycle then repeats.